

Prepared By	08-Aug-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	08-Aug-2012	Mr.Chatchawan S.		
Approved By	08-Aug-2012	Mr.Amnauy W.		
Certified By	08-Aug-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	08-Aug-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-020-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-011**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No **RNC-WPS-011**

Date **6-Aug-12**

Revision No **0**

Supporting PQR No.(s) **RNC-PQR-007**

Date **13-Jun-12**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic **Manual** Machine, or Semi-Auto)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) **(No)** **X**

Backing Material (Type) **N A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No **8** Group No **1** to P-No **8** Group No **1**

Material Specification Type and Grade **A 312 Gr. TP 304, TP 304L, A182 F304, F304L** TO **A 312 Gr. TP 304, TP 304L, A182 F304, F304L**

Chem Analysis and Mech Prop

Thickness Range

Base Metal Groove **1.6 mm - 7.82 mm** Fillet **All**

Pipe Dia Range Groove **All** Fillet **All**

Other

FILLER METALS (QW-404)

Spec No (SFA) **5.9**

AWS No. (Class) **ER-308L**

F-No **6**

A-No **8**

Size of Filler Metals **Ø 2.0 & 2.4 mm**

Deposited Weld Metals **3.91 mm**

Thickness Range :

Groove **7.82 mm Max.**

Fillet **All**

Electrode-Flux (Class) **-**

Flux Trade Name **-**

Consumable Insert

Other **Kobelco TGS-308L or Equivalents**

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression **Up** **X** **Down**

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N A**

Time Range **N A**

Notes

PREHEAT (QW-406)

Preheat Temp Min **101°C**

Interpass Temp Max **150°C**

Preheat Maintenance **N A**

Notes

GAS (QW-408)

Percent Composition

Gas(es)	(Mixture)	Flow Rate
Shielding Argon	99.99%	12-16 Ltr/Min
Trailing -	-	-
Backing Argon	99.99%	13-17 Ltr/Min

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-011 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 70-130 Volts (Range) 8-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

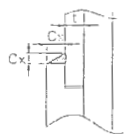
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5-11 Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER-308L	2.4	DC EN	80-110	8-12	5-9	
2nd	GTAW	ER-308L	2.4	DC EN	100-130	10-13	7-11	
Other(s)	GTAW	ER-308L	2.4	DC EN	70-120	10-13	6-9	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME IX, 2007 latest Revision.

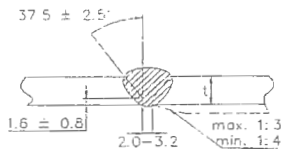
Prepared by: Mv. Chatchaman.s Signed: [Signature] Date: 6/08/12
 Reviewed by: Mv. Chatchaman.s Signed: [Signature] Date: 6/08/12
 Approved by: A. ARUNAY W. Signed: [Signature] Date: 15/08/12

ATTACHED FIGURE JOINT SKETCH



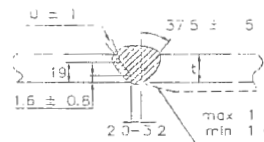
C_x (min.) = $1.25t$ but not less than 3.2 mm

JOINT No. A1



Misalignment (max.) 1.5 mm.

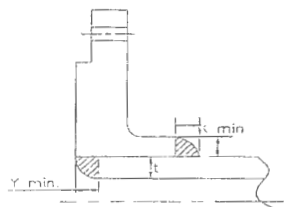
JOINT No. B1 & B2



Approx. 1.6 mm

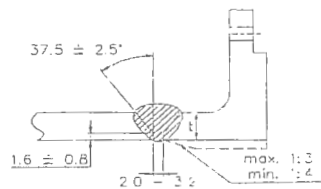
X min = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



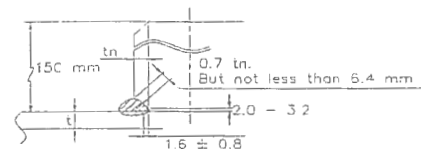
X min. = The lesser of $1.4t$ or thickness of the hub.
 Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



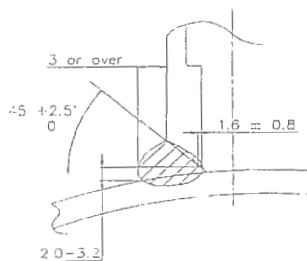
Misalignment (max.) = 1.5 mm

JOINT No. B3

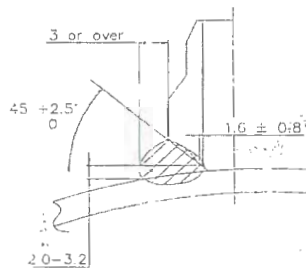


Misalignment (max.) = The lesser of 3.2 mm. or $0.5t_n$

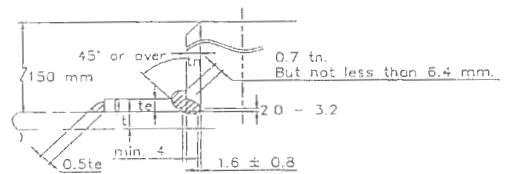
JOINT No. B4



JOINT No. C1

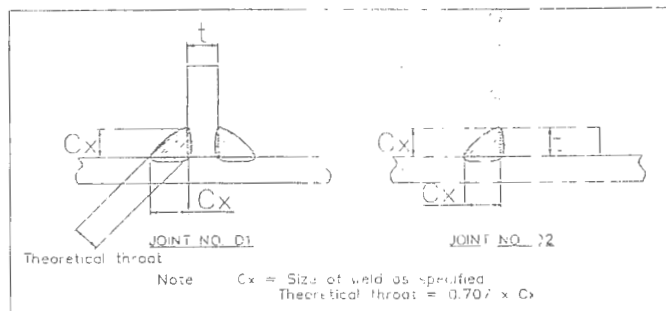


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5t_n$

JOINT No. B5

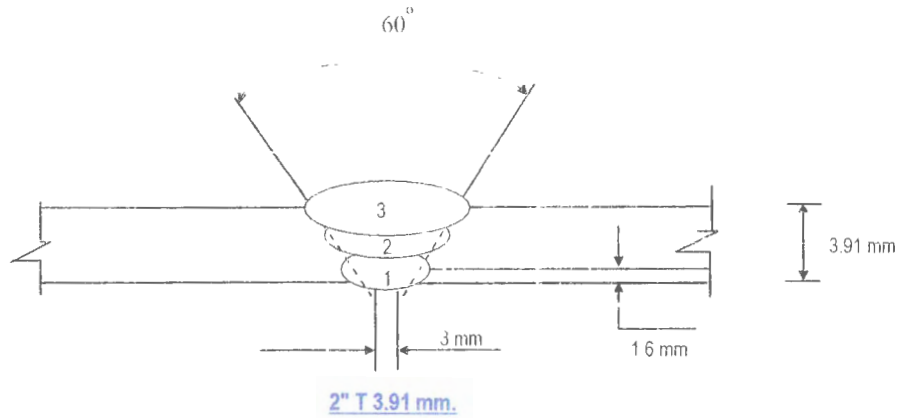


JOINT NO. D1

JOINT NO. D2

Note C_x = Size of weld as specified
Theoretical throat = $0.707 \times C_x$

Welding Sequences for PQR-RNC-007 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Wire Feed		Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER316L	2.4	55	10	DCEN			6	8.40	15	16
2	GTAW	ER316L	2.4	60	10	DCEN			9	7.06	15	16
3	GTAW	ER316L	2.4	60	10	DCEN			8	6.52	15	16

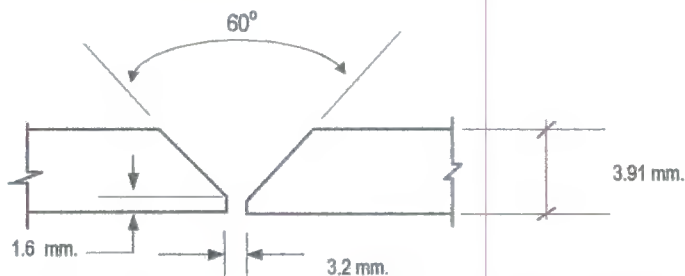
Procedure Qualification Record (PQR)

ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : **RNC-PQR - 007**

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-007	Date :	13 June 2012
WPS Number :	RNC-WPS-007		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A 312	To	A312	Temperature	N/A		
Type or Grade	TP316L	To	TP316L	Hold Time	N/A		
P. No.	P. 8	To	P. 8	Others	N/A		
Thickness of Test Coupon	3.91 mm.			GAS (QW-408)			
Diameter of Test Coupon	Ø 2"						
Others							
FILLER METALS (QW-404)				Percent Composition			
SFA Specification	A 5.9				Gas (es)	Mixture	Flow Rate
AWS Classification	ER316L			Shielding	Argon	99.95%	15 L/Min.
Filler Metal F-No.	6			Trailing	-	-	-
Weld Metal Analysis A-No.	8			Backing	Argon	99.95%	16 L/Min.
Size of Filler Metal	Ø 2.4 mm			ELECTRICAL CHARACTERISTICS (QW-409)			
Others	Kobelco (TGS-316L)			Current	DC		
Weld Metal Thickness	-			Polarity	EN		
POSITION (QW-405)				Amperage	84 - 106	Voltage	10
Position of Groove	6G			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Progression of Welding (Uphill or Down)			Up Hill	Others	N/A		
Others	N/A			TECHNIQUE (QW-410)			
PREHEAT (QW-406)				Travel Speed	6 - 9 cm/min.		
Preheat Temperature	Ambient Temperature			String or Weave Bead	Both		
Interpass Temperature (max.)	144° C			Oscillation	N/A		
Others	N/A			Multipass or single Pass	Multiple Pass		
				Multipass or Single Electrodes	Single Electrode		
				Others	N/A		

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number: **PQR-RNC-007**

Tensile Test Report No.TS-12-06-107

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2368/12 TR1	12.43	4.06	50.466	28.753	569.754	WELD ZONE
2368/12 TR2	13.12	4.08	53.530	30.165	563.524	WELD ZONE

Guided Bend Tests (QW-160) Report No.BD-12-06-105

Type and Figure Number	Type of Bend	Indication	Results
2368/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2368/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2368/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2368/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR -003
Hardness Testing Report No. HN-12-06-082
Macrostructure Report No. MA-12-06-104
Impact Testing Report No.IM-12-06-140

Welders Name Mr.Boontham J. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : 
Date : 29.06.12

Inspected by : Mr. N. Bhavut M.
Certified by : Mr. O. Chanthiet R.
Date : 10 



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T TABMA A MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX (038) 633 477 MOBILE (086) 340 9978

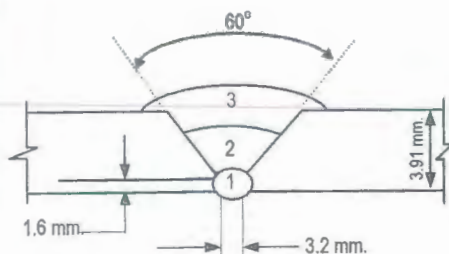
Procedure Qualification Record (PQR)

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-003			
		Customer Order No. : N/A		WPS No. : RNC-WPS-007		Date : 13-Jun-12			
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.BOONTHAM J.	GTAW	6G	ASME IX	A312TP316L	ø 2" Thk. 3.91 mm.	ER-316L	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER-316L	2.4	DCEN	84	10	6	140	8.40	Argon	15	Argon	16
2	GTAW	ER-316L	2.4	DCEN	106	10	9	142	7.06	Argon	15	Argon	16
3	GTAW	ER-316L	2.4	DCEN	87	10	8	144	6.52	Argon	15	Argon	16

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

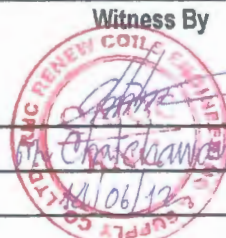
DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

A
ANNUAY W.
29-06-12

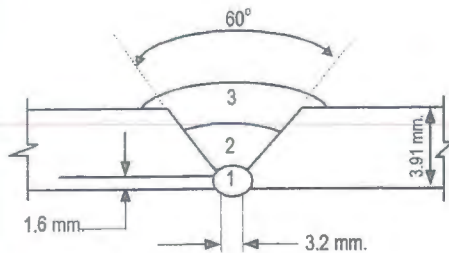
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-003 A
Customer Order No. : N/A	WPS No. : RNC-WPS-007	Date : 29-Jun-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.BOONTHAM J.	GTAW	6G	ASME IX	A312TP316L	ø 2" Thk. 3.91 mm.	ER-316L	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER-316L	2.4	DCEN	85	11	6	138	9.35	Argon	15	Argon	12
2	GTAW	ER-316L	2.4	DCEN	95	12	7	142	9.77	Argon	15	Argon	12
3	GTAW	ER-316L	2.4	DCEN	95	12	7	145	9.77	Argon	15	Argon	12

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

PAE
TECHNICAL SERVICE
30-Jun-2012

SIGNATURE :

NAME :

DATE :

Chatchand S
20/06/12

SIGNATURE :

NAME :

DATE :

ANNUAY W.
29-06-12



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Radiographic Examination Report

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]

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Renew Coils Engineering & Supply CO., LTD.

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TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Mechanical Test Report



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th



22688

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-107
Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Test	Reduced Section Tension
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2368/12 TR1	N/A	4.06	12.43	50.466	N/A	N/A	N/A	28.753	569.754	N/A	WELD ZONE
2	2368/12 TR2	N/A	4.08	13.12	53.530	N/A	N/A	N/A	30.165	563.524	N/A	WELD ZONE

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-007

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 9 / 07 / 12



Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 9 / 07 / 12

LPN PLATE MILL

- Notes**
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Testing Laboratory LPN Plate Mill Public Company Limited

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22689

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. BD-12-06-105
Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Diameter of Mandrel	16 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2368/12 BF1	Transverse face bend	3.89x19.05x157.47 mm	No Open Discontinuity	N/A	-
2	2368/12 BF2	Transverse face bend	4.03x19.04x160.14 mm	No Open Discontinuity	N/A	-
3	2368/12 BR1	Transverse root bend	4.10x19.11x159.87 mm	No Open Discontinuity	N/A	-
4	2368/12 BR2	Transverse root bend	3.94x19.05x159.34 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-007
Welding Process : GTAW
Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by : 
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date 05 / 07 / 12



Approved by : 
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date 05 / 07 / 12

LPN PLATE MILL

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



22948

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-140

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{sl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A312 TP316L	Type of Test	Charpy V-Notch Impact
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2368/12-1	-20°C	2.519	10.014	54.72	33.82	32.18	Base Metal	-
2	2368/12-2	-20°C	2.515	10.011	54.71	30.35			
3	2368/12-3	-20°C	2.518	10.013	54.72	32.38			
4	2368/12-1	-20°C	2.504	10.014	54.83	32.17	31.42	Weld Metal	-
5	2368/12-2	-20°C	2.504	10.001	54.86	32.37			
6	2368/12-3	-20°C	2.504	10.001	54.82	29.73			
7	2368/12-1	-20°C	2.501	9.999	54.83	36.17	37.44	HAZ	-
8	2368/12-2	-20°C	2.504	10.000	54.87	37.92			
9	2368/12-3	-20°C	2.500	10.004	54.95	38.24			

Additional details : WPS No. RNC-WPS-007

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-316L/TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 13 / 07 / 12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 13 / 07 / 12

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

1111

Macrostructure Analysis Report

Page 1 of 1

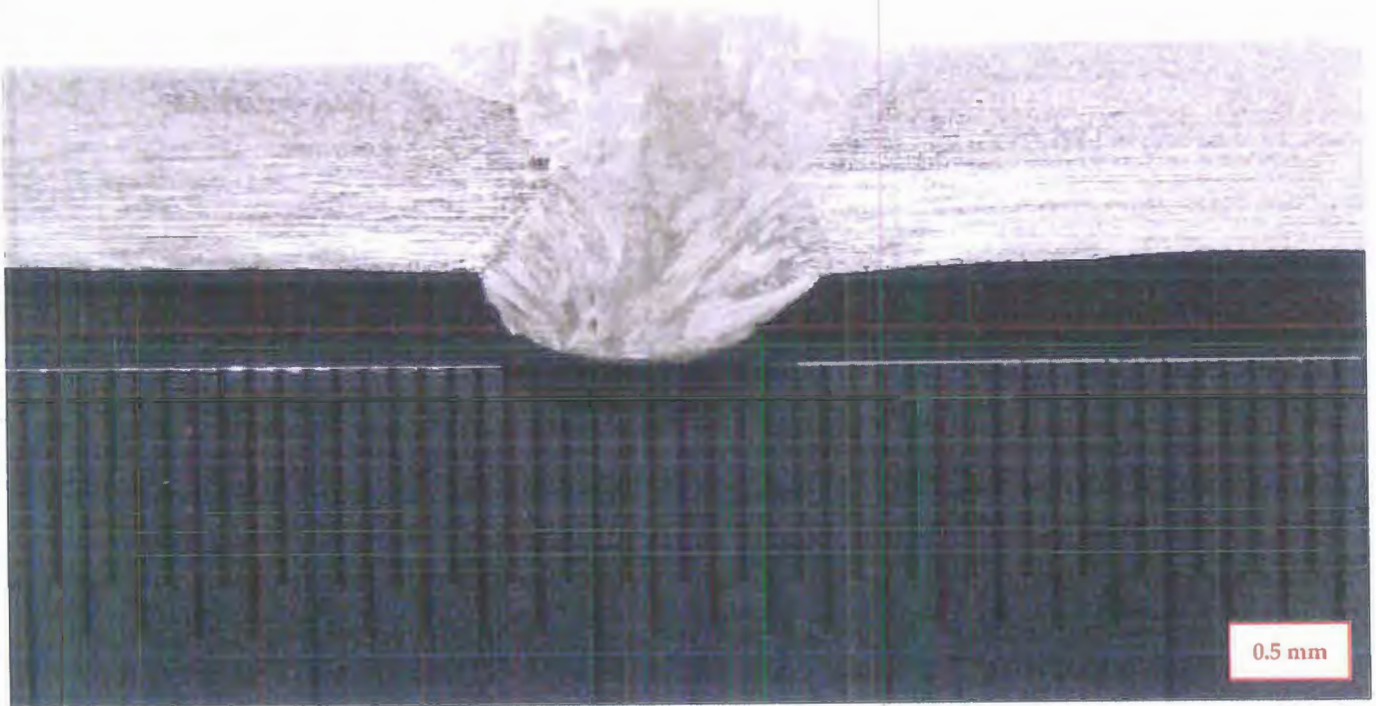
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-104

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Test No.	2368/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A312TP316L	Temperature	28 °C
Dimension	Ø2"Sch40(3.91 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
Filler/Electrode Spec't : ER-316L TGS-316L

Reported by :

(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 27/7/12



Approved by :

(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 27/7/12

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

12470

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-082

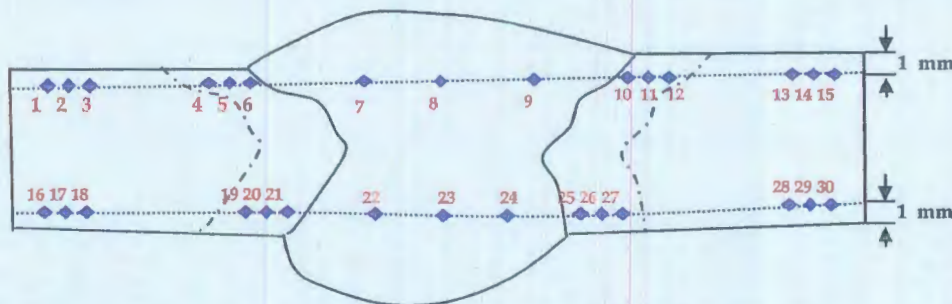
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-007	Material Characterization	Pipe
Test No.	2368/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A312TP316L	Equipment	Mitutoyo, HV-115
Dimension	Ø2"Sch40(3.91 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	173.9	16	BASE	173.0	*****		
2	BASE	173.6	17	BASE	172.8			
3	BASE	174.2	18	BASE	172.3			
4	HAZ	169.2	19	HAZ	179.8			
5	HAZ	163.5	20	HAZ	173.6			
6	HAZ	159.3	21	HAZ	160.5			
7	WELD	149.7	22	WELD	146.7			
8	WELD	154.1	23	WELD	154.8			
9	WELD	152.1	24	WELD	149.4			
10	HAZ	156.2	25	HAZ	160.7			
11	HAZ	154.7	26	HAZ	167.7			
12	HAZ	160.3	27	HAZ	158.3			
13	BASE	165.4	28	BASE	159.5			
14	BASE	166.1	29	BASE	160.1			
15	BASE	166.0	30	BASE	161.3			

Additional details : Filler/Electrode Spec't : ER-316L TGS-316L

Reported by :

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date : 23/07/12



Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date : 23/07/12

LPN PLATE MILL

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Consumable Certificate

TKW120106-KOBE

INSPECTION CERTIFICATE
(検 査 証 明 書)PURCHASER
需要家WELDING ROD OR WIRE
溶加棒又はワイヤCERTIFICATE No
証明書番号: 220120244
DATE OF ISSUE
発行日: 2012.03.21

TRADE DESIGNATION 品 名 (型 番)	DIMENSION 寸 法 (mm)	MFG.No 製造番号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類
TG-S316L	(2.4) mm 3/32 inch	E36762	AWS A5.9 ER316L ASME SFA-5.9 ER316L

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	CR	MO	NB	N	FN	FS
ROD OR WIRE 棒又はワイヤ	0.008	0.40	1.55	0.024	0.002	0.15	12.18	18.58	2.14	<0.01	0.02	10	6
ELEMENTS 成分	FNW												
ROD OR WIRE 棒又はワイヤ	9												

FS = FERRITE % (SCHAEFFLER DIAGRAM)

FN = FERRITE NUMBER (DELONG DIAGRAM)

FNW= FERRITE NUMBER (WRC)

TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能				HARDNESS PROPERTIES 硬さ	
YIELD POINT 降 伏 点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引 張 強 さ	ELONGATION 伸 び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸 収 エ ネ ル ギ ー				
(-----) N/mm ² ----- ksi	(-) N/mm ² - ksi	(-----) N/mm ² ----- ksi	- %	(-) °C - °F	AVG. 平均 (-) J - ft-lbf	EACH ()			-

WELDING CONDITIONS 溶 接 条 件				POSTWELD HEAT TREATMENT 溶 接 後 熱 処 理	
TYPE OF CURRENT 電 流 の 種 類	-	SHIELDING GAS シールドガス	-		
AMPERAGE 溶 接 電 流	- A			(-) °C × - °F ×	- h - h
ARC VOLTAGE アーク電圧	- V				

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。◆KOBEL STEEL, LTD.
WELDING BUSINESS FUJISAWA WORKS
株式会社 神戸製鋼所 溶接事業部門 藤沢事業所APPROVED BY CHIEF INSPECTOR
主任検査員 承認済

V. Cerachi



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Material Certificate

CERTIFIED MATERIAL TEST REPORT

E3

Messrs. OR MENA LTD

Date Apr. 20, 2011

Note Grade TP 316/316L

Sheet No. 627556

Condition Solution Treated

Contract No. T13TH26

Heat No. 801971

Packing List No. ET3827

Size 2 X SCH 40.S

Bundle No. 3 - 4

Length 6,000 - 6,003 mm

SANYO SPECIAL STEEL CO., LTD.

Pieces 35

Quality Assurance Department

Quality Certification Group

3007, Nakashima, Shikama-ku, Himeji, Japan

TEL (079) 235-6029

FAX (079) 235-6227

Chemical Composition (%)	C	Si	Mn	P	S	Ni	Cr	Mo
	X1000	X100	X100	X1000	X1000	X100	X100	X100
Standard	35	100	200	45	30	1100	1600	200
	max	max	max	max	max	1400	1800	300
Cast	14	46	127	36	4	1123	1626	206

Mechanical Test	Type of Specimen	0.2% Yield Strength	Tensile Strength	Elongation
From Standard Product		30 min Ksi	75 min Ksi	35 min %
Result	ASTM	43	80	50

Corrosion Test (ASTM A262-E) : GOOD

Flattening Test GOOD

Dimension Inspection GOOD

Spark Test GOOD

Eddy Current Test GOOD

Specification ASTM A312
ASME SA312

Remarks Heat Treatment : 1070°C X 13M W.C
SEAMLESS STAINLESS STEEL PIPES
HARDNESS ACC. TO NACE MR0175:GOOD
PMI is based on Spark Test
DIN3.1 according to DIN EN 10204 (DIN 50049)

END OF REPORT



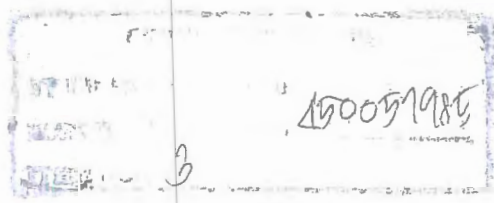
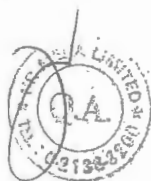
SS 316
CR 16.26

NI 11.23

MM 1.27

MS 2.06

per spec
20/4/11



We hereby certify that the material described herein has been made in accordance with the rules of the contract.

Head of Quality Certification Group

K. Hashizume

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-007-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RE-WS-01**

Sheet 1 of 2

WELDING PROCEDURE SPECIFICATIONS (WPS)

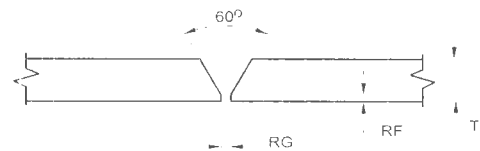
Company Name RENEW COILS ENGINEERING & SUPPLY CO., LTD.
 Welding Procedure Specification no. RE-WS-01
 Revision no. 0 Date -
 Welding Process(es) 1/ GTAW 2/ -
 Impact Test Requirement ☐ Yes ☒ No

By Mr. Amnuay Wungnai (Sign/Date)
 Date FEBRUARY 17, 2007
 Supporting PQR no.(s) RE-PR-01
 Type(s) 1/ Manual
 2/ -

JOINTS (QW-402)

Joint Design Butt joint (Single-V groove)
 Backing ☐ Yes ☒ No
 Backing Material (Type) 1/ -
 2/ -
☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other
 Applicable of Tack Welding ☒ Yes ☐ No
 Applicable of Repair Welding ☒ Yes ☐ No
 Retainer None

Details



RG = 0 - 3.0 mm.
 RF = 0 - 2.0 mm.
 T = 1.5 - 6.0 mm.

BASE METALS (QW-403)

P-No. 43 Group no. - to P-No. 43 Group no. -
 OR Specification Type and Grade SB 575 N06200 to SB 575 N06200
 OR Chem. Analysis and Mech. Prop. NA to NA
 Base Metal Thk. Range (T, mm.) Groove 1.5 - 6.0 mm. Fillet All thicknesses
 Pipe Dia. Range Groove All diameters Fillet All diameters
 Other NA

FILLER METALS (QW-404)

Spec. no. (SFA) 1/ 5.14 2/ -
 AWS no. (Class) 1/ ER NiCrMo-17 2/ -
 F-no. 1/ 43 2/ -
 A-no. 1/ See attach sheet 2/ -
 Size of Filler Metals 1/ Ø 1.6, 2.4 mm. 2/ -
 Deposited Weld Metal Thk. Range (t, mm.)
 Groove 1/ 6.0 mm.(Max.) 2/ -
 Fillet 1/ Unlimited 2/ -
 Type-wire 1/ NA 2/ -
 Electrode-Flux (class) NA
 Flux Trade Name NA
 Consumable Insert None
 Other NA

WPS no. RE-WS-01 Rev. 0**POSITIONS (QW-405)**

Position of Groove 1/ All 2/ All
 Welding Progression : Up hill Down -
 Position(s) of Fillet 1/ All 2/ All

POSTWELD Mr. Amnuay Wungnai

Temperature Range (NO PWHT) °C
 Time Range NA hr.
 Other None

PREHEAT (QW-406)

Preheat Temp. Min. 10 °C
 Interpass Temp. Max. 93 °C
 Preheat Maintenance Cool in still air
 Other NA

GAS (QW-408)

Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Ar ₂	Welding grade	12-17 l/min
Trailing	NA	-	-
Backing	NA	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current AC or DC 1/ DC 2/ - Polarity 1/ EN 2/ -
 Amps (Range) See table below Volts (Range) See table below
 Tungsten Electrode Size and Type Ø 2.4 mm ; 2% Thoriated (EWTh-2)
 Mode of Metal Transfer for GMAW NA
 Electrode Wire Feed Speed Range NA

TECHNIQUE (QW-410)

String or Weave Bead 1/ String or Weave 2/ -
 Orifice or Gas Cup Size No. 4 - 8
 Initial and Interpass Cleaning Wire brushing and/or Grinding
 Method of Back Gouging Grinding
 Oscillation NA
 Contact Tube to Work Distance 1/ NA 2/ -
 Multiple or Single Pass (per side) 1/ Multiple 2/ -
 Multiple or Single Electrodes 1/ Single 2/ -
 Travel Speed (Range) See Table below
 Peening NA
 Other NA

Weld Layer(s)	Process(es)	Filler Metal		Current		Volt Range	Travel Speed Range: cm/minute	Other Flow rate
		Class	Dia.	Type Polar.	Amp. Range			
All	GTAW	ERNiCrMo-17	1.6, 2.4 mm.	DCEN	50-115	9-12	5-12	Ar ₂ 12-17 l/min

RENEW COILS ENGINEERING & SUPPLY CO., LTD.

RNC

8 Tesabambumrung Rd., Tambol Mabtaput, Amphur Muang, Rayong, Thailand 21150

Tel. 038-681-237, Fax. 038-681-247

Sheet 1 of 3

PROCEDURE QUALIFICATION RECORD (PQR)

Company Name RENEW COILS ENGINEERING & SUPPLY CO., LTD.Procedure Qualification Record no. RE-PR-01 Date FEBRUARY 17, 2007WPS no. RE-WS-01 Revision. no. 0Welding Process(es) 1/ GTAW 2/ -Types (Manual, Automatic, Semi-Automatic) 1/ Manual 2/ -Impact Test Requirement ☐ Yes ☒ No

JOINTS (QW-402)

SEE PAGE 3 OF 3

BASE METALS (QW-403)

Material Spec. SB 575 to SB 575
 Type of Grade - to -
 P-no. 43 to P-no. 43
 Thickness of Test Coupon, T NA mm.
 Diameter of Test Coupon O.D. 2"
 Other NA

POSTWELD HEAT TREATMENT (QW-407)

Temperature (NO PWHT) °C
 Time NA hr.
 Other None

GAS (QW-408)

	Percent Composition		
	Gas(es)	Mixture	Flow Rate
Shielding	Ar ₂	100%	17 l/min
Trailing	NA	-	-
Backing	NA	-	-

FILLER METALS (QW-404)

SFA Specification 1/ 5.14 2/ -
 AWS Classification 1/ ERNiCrMo-17 2/ -
 Filler Metal F.no. 1/ 43 2/ -
 Weld Metal Analysis A. No. 1/ See attach sheet
 Size of Filler Metal 1/ Ø 2.4 mm. 2/ -
 Other NA
 Deposited Weld Metal 1/ 3.0 mm.
 thickness, t (mm.) 2/ -

ELECTRICAL CHARACTERISTICS (QW-409)

Current 1/ DC 2/ -
 Polarity 1/ EN 2/ -
 Amps. 1/ See sheet 3 2/ -
 Volts 1/ See sheet 3 2/ -
 Tungsten Electrode Size Ø 2.4 mm
 Other 2% Thoriated (EWTh-2)

POSITIONS (QW-405)

Position of Groove 1/ 6G 2/ -
 Welding Progression (Uphill, Downhill)
 1/ Uphill 2/ -
 Other None

TECHNIQUE (QW-410)

Travel Speed 1/ See sheet 3 2/ -
 String or Weave Bead 1/ Weave 2/ -
 Oscillation NA
 Multipass or Single Pass (per side) Multiple
 Single or Multiple Electrodes Single
 Other NA

PREHEAT (QW-406)

Preheat Temp. Ambient (~30) °C
 Interpass Temp. 80 °C
 Other Cool in still air

RENEW COILS ENGINEERING & SUPPLY CO., LTD.

RNC

8 Tesabambumrung Rd., Tambol Mabtaput, Amphur Muang, Rayong, Thailand 21150

Tel. 038-681-237, Fax. 038-681-247

Sheet 2 of 3

PQR Back

PQR no. RE-PR-01

TENSILE TEST (QW-150)

Specimen no.	Width (mm.)	Thickness (mm.)	Area (mm ²)	Ultimate Tensile Load (N)	Ultimate Unit Stress (N/mm ²)	Character of Failure and Location
TS 88 / 2007	13.16	3.30	43.43	34,900	803.59	At weld
TS 89 / 2007	13.14	3.30	43.36	32,500	749.54	At weld

GUIDED BEND TEST (QW-160)

Specimen no.	Type & Figure No. of Bend Tests	Results	Remarks
FB 207 / 2007	Transverse Face Bend (QW 462.3 (a))	Acceptable	No Discontinuity
RB 208 / 2007	Transverse Root Bend (QW 462.3(a))	Acceptable	No Discontinuity
FB 209 / 2007	Transverse Face Bend (QW 462.3 (a))	Acceptable	No Discontinuity
RB 210 / 2007	Transverse Root Bend (QW 462.3(a))	Acceptable	No Discontinuity

TOUGHNESS TEST (QW-170)

Specimen no.	Notch Location	Notch Type	Test Temp. °C	Impact Value			Drop Weight Break	
				ft.-lbs.	%Shear	Mils	Break	No Break
N/A	-	-	-	-	-	-	-	-
N/A	-	-	-	-	-	-	-	-
N/A	-	-	-	-	-	-	-	-
N/A	-	-	-	-	-	-	-	-
N/A	-	-	-	-	-	-	-	-
N/A	-	-	-	-	-	-	-	-

FILLET-WELD TEST (QW-180)

Results - Satisfactory : ☐ Yes ☐ No Penetration into Parent Metal : ☐ Yes ☐ No

Macro - Results : NA

OTHER TESTS

Type of Test RADIOGRAPHIC TEST REPORT NO. R07-0896-1-P42

Deposit Analysis NA

Other None

Welder's name Mr. Suriyon Sanitnoi Clock no. - Stamp no. W-005

Test conducted by SIWA TESTING INSPECTION & CONSULTING CO.,LTD. Laboratory Test no. R07-0896-2,3-P42

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.

Manufacturer RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Date Certify



บริษัท ซีว: เทสติ้ง อินสเปคชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

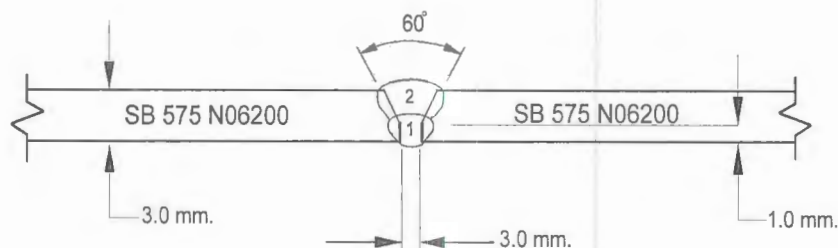
Sheet 3 of 3

Actual Records of Welding Variable

ACC. TO ASME SECTION IX WELDING VARIABLE

WPS NO. RE-PR-01 Revision No. 0

Front side



Back side

Backing ☐ Yes ☒ No
Retainers ☐ Yes ☒ No

PASS SEQUENCE

Side	Pass	Process	Electrode or Filler		Amps./ Polarity	Volts	Travel Speed (cm./minute)	Remarks (gas flow rate)
			Type	Size				
Front	1	GTAW	ERNiCrMO-17	Ø 2.4 mm.	50 /DCEN	10 V.	6	Ar ₂ 17 l/min
	2	GTAW	ERNiCrMO-17	Ø 2.4 mm.	71 /DCEN	11 V.	10	Ar ₂ 17 l/min

Witnessed by :

Ch. Chirot
(Mr. Chirot Ch.)

Date : FEBRUARY 17, 2007

01-15-07/C9:24AM;

ACCOUNTING

7/139374596

37 4

HAYNES

Wire Company

SOLD TO

Haynes Int. - Houston
12231 FM 329
Northwoods Industrial Park
Houston, TX 7041
USA
Attn: QA

159 N. Egerton Rd
P.O. Box 577
Mountain Home, NC 28758
828-692-5791
828-697-9818 FAX

PRODUCT CERTIFICATION

WORK ORDER
S01583

HEAT NUMBER
2316-6-8098

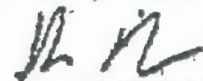
SALES ORDER / RLS
00124817

ISO 9001:2000 Certified
Certificate #: FM70739

CUSTOMER P.O. 3326/SO# 482207	CUSTOMER PART	QUANTITY 602 Lbs	COILS LADING NO 11 00001382	SHIPMENT DATE 12/28/2006
SPECIFICATION C2000-0930-10TT-M00 HASTELLOY® C-2000™ Alloy 0.0930 diameter, 36", 38" TIG Tube, Matte, 10.0 lbs./Package				
CERTIFICATION REQUIREMENTS PS-6082-4 Class A AWS A5.14-05 ERNiCrMo-17 SME-SFA-5.14-04 ERNiCrMo-17 EN10204 3.1				
Chemical				
Al 0.0015	C 0.0014	Co 0.1436	Cr 22.3751	Cu 1.4692
Fe 1.134	Mn 0.2271	Mo 15.5339	Ni BAL	P 0.004
S 0.0023	Si 0.02			
Mechanical by Lot				
TEST Diameter	UNITS In	HIGH 0.0928	LOW 0.0927	AVERAGE 0.0928
TEST Alloy ID	RESULT Pass			
Surface Check	Pass			
Visual and dimensional examination was satisfactory. Material, when shipped, is free from contamination by mercury, radium, alpha source, and low melting point elements.				
End of Certification				

This is to certify that all required samplings, inspections, and tests have been performed in accordance with the specification requirements. The test report represents the actual attributes of the material furnished and the values shown are correct and true. The material described by this certificate is in full compliance with all order and inspection requirements. We hereby certify that the above data are in accordance with the specification requirements.

John Cunningham
QA Representative



Date Printed 12/28/2006

01-15-07; 09:24AM;

ACCOUNTING

77139374596

3/ 4

HAYNES

Wire Company

SOLD TO

Haynes Int. - Houston
12231 FM 529
Northwoods Industrial Park
Houston, TX 7041
USA
Attn: QA

158 N. Egerton Rd
P.O. Box 577
Mountain Home, NC 28758
828-692-5791
828-697-9615 FAX

PRODUCT CERTIFICATION

WORK ORDER
S01583

HEAT NUMBER
2315-6-8098

SALES ORDER / RLS
00124817

ISO 9001:2000 Certified
Certificate #: FM70739

CUSTOMER P.O. 3325/SO# 482207	CUSTOMER PART	QUANTITY 602 Lbs	COILS 11	LADING NO 00001382	SHIPMENT DATE 12/28/2006
SPECIFICATION C2000-0930-10TT-M00 HASTELLOY® C-2000™ Alloy 0.0930 diameter, 36", 38" TIG Tube, Matte, 10.0 lbs./Package					
CERTIFICATION REQUIREMENTS PS-6082-4 Class A AWS A5.14-05 ERNiCrMo-17 SME-SFA-5.14-04 ERNiCrMo-17 EN10204 3.1					
Chemical					
Al 0.0015	C 0.0014	Co 0.1436	Cr 22.3751	Cu 1.4692	
Fe 1.134	Mn 0.2271	Mo 15.5339	Ni BAL	P 0.004	
S 0.0023	Si 0.02				
Mechanical by Lot					
TEST Diameter	UNITS In	HIGH 0.0928	LOW 0.0927	AVERAGE 0.0928	
TEST Alloy ID	RESULT Pass				
Surface Check	Pass				
Visual and dimensional examination was satisfactory. Material, when shipped, is free from contamination by mercury, radium, alpha source, and low melting point elements.					
End of Certification					

This is to certify that all required samplings, inspections, and tests have been performed in accordance with the specification requirements. The test report represents the actual attributes of the material furnished and the values shown are correct and true. The material described by this certificate is in full compliance with all order and inspection requirements. We hereby certify that the above data are in accordance with the specification requirements.

John Cunningham
QA Representative



Date Printed 12/28/2006

SUMMARY OF WELDER/WELDING OPERATOR PERFORMANCE QUALIFICATION

Manufacturer :	RENEW COILS ENGINEERING & SUPPLY CO., LTD.		
Project :	WPS/PQR		
Applicable Code :	ASME SEC. IX	Edition:	2004 Addenda: -
Witness Date :	FEBRUARY 17, 2007	Place :	RAYONG
Material Specification:	SB 575 N06200		
Electrode/Filler Metal Classification :	ERNiCrMo-17		

[illegible]

Legend : RT = Radiographic Test, NA = not applicable, ACC = Accept, REJ = Reject

Note. For complete qualification range, see attached certificate(s).



RADIOGRAPHIC TEST REPORT

Page 1 of 1

Report No. :	R07-0896-1-P42
Test Date :	February 20, 2007
Client :	Renew Coils Engineering & Supply Co., Ltd.
Description of Work :	PQR
Place of Inspection :	STIC
Specification :	ASME SEC. IX
Wall Thickness :	Ø 2" X 3.0 mm., GTAW
Source :	Gamma Ray
Technique :	Double Wall - Double Image
Film Type :	Kodak AA
Total Films :	3.5" x 8.5" = 2 Film(s), 3.5" x 17" = - Film(s)

[illegible]



บริษัท สิวา: เทรดดิ้ง อินสเปคชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

Welder Cert. no. 013/2007



Company's name : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Welder's name : Mr. Suriyon Sanitnoi

Identification no. W-005

Test Description

Identification of WPS followed or joint type RE-WS-01 ☒ Test coupon ☐ Production Weld
Specification of Base Metal(s) SB 575 N06200 Thickness 3.0 mm.
Filler metal or electrode specification (SFA) 5.14 Classification(s) ERNiCrMo-17

Testing Conditions and Qualification Limits

Welding Variables (QW-350)

Welding process(es) used
Type (ie; manual, semi-auto) used
Backing (metal, weld metal, welded from both sides, flux, etc.) (QW-402)
Base metal (P-or S-Number) 43 to (P-or S-Number)(QW-403)

☐ Plate ☒ Pipe (enter diameter, if pipe)

Groove
Fillet

Filler metal F-No.(QW-404)

Filler metal product form (solid/cored/flux-cored) GTAW/PAW (QW-404)

Consumable insert for GTAW or PAW (QW-404)

Weld deposit thickness for each welding process (QW-404)

Process 1: GTAW 3 layers minimum ☐ Yes ☒ No
Process 2: - 3 layers minimum ☐ Yes ☐ No

Welding position (QW-405)

Groove weld test

Plate & Pipe O.D. > 24"(610 mm)

Pipe O.D. ≤ 24"(610 mm)

Fillet weld test

Vertical Progression (uphill or downhill) (QW-405)

Backing gas for GTAW, PAW or GMAW (QW-408)

GMAW transfer Mode (QW-409)

GTAW welding current type/polarity (QW-409)

Actual Values

GTAW

Manual

Welding without backing

43

O.D. 2"

NA

6

Solid Wire

None

3.0 mm.

NA

6G

NA

Uphill

Argon : 100 %

NA

DC/EN

Range Qualified

GTAW

Manual

With and/or without backing

P-or S-No. 34, and

P-or S-No. 41 Thru P-or S-No. 49

O.D. 1" to Unlimited

All thicknesses & diameters

With or without backing : 6

With solid filler metal

6.0 mm.(Max.)&all fillet sizes

-

All position

All position

All position

Uphill

Argon : 100 %

-

DC/EN

RESULTS

Visual examination of Completed Weld (QW-302.4) ACCEPTABLE

Guided Bend Test Result

☐ Transverse root and face [QW-462.3(a)];

☐ Longitudinal root and face [QW-462.3(b)];

☐ Side [QW-462.2)

Type and Specimen no.	Results
None	-
None	-

Type and Specimen no.	Results
None	-
None	-

Radiographic Test Results (QW-191.2)

(For alternative qualification of groove welds by radiography)

Radiographic test result (QW-304 and QW-305) ACCEPTABLE

Test conducted by SIWA TESTING INSPECTION & CONSULTING CO.,LTD. Report no. R07-0896-1-P42

Fillet Weld Test Results (See QW-462.4(b),(c))

Fillet weld - Fracture test NA Length and percent of defects -
Macro test fusion NA Fillet leg size (in.) - X Concavity/convexity(in.) -
Mechanical tests conducted by - Report no. -

Welding Test Witnessed / Reviewed by : *Ch. Chirot* / *K. Preecha* (Mr. Chirot Ch. / Mr. Preecha K.) Witness date : FEBRUARY 17, 2007

SIWA TESTING INSPECTION & CONSULTING CO., LTD.

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of the ASME Section IX. Current Edition, Addenda.

Certified by : (Manufacturer's representative)

Date :



TENSILE TEST REPORT

Report no. : R07-0896-2-P42 Page 1 of 1

Test Date : February 26, 2007

Client : RENEW COILS ENGINEERING & SUPPLY CO., LTD.

Tested Product : PQR. No. RE-PR-01 (Welded Pipe Ø2")

Material : SB 575 N06022

Project : -

Applicable Standard : ASME Sec. IX : 2004

Thickness : 3.0 mm

No.	Specimen No.	Size of Specimen (mm)	Gauge Length (mm)	Elongation (mm)	Elongation (%)	Tested Area (mm ²)	Load		Stress		Remark
							Yield (N)	Ultimate (N)	Yield (N/mm ²)	Ultimate (N/mm ²)	
1	TS 88 / 2007	13.16 x 3.30	-	-	-	43.43	-	34,900	-	803.59	FAW - Accept
2	TS 89 / 2007	13.14 x 3.30	-	-	-	43.36	-	32,500	-	749.54	

Remark : FAW - Failure at weld

FOOW - Failure out of weld

Total	2	Specimen(s)	☐ Attached Report =	Page(s)
Operated by	Nopporn Karnjanajutaphan		Reviewed and Supervised by :	
Certified by	 (Patoomporn Phongchotip / Project Engineer)		Date : 05 MAR 2007	

GUIDED BEND TEST REPORT

Test Date : February 26, 2007

Specimen No.	Type of Bending	Size of Specimen	Fracture Location	Remark
FB 207 / 2007	Transverse Face Bend	19.0 x 3.0	No Discontinuity	Accept
RB 208 / 2007	Transverse Root Bend	19.0 x 3.0	No Discontinuity	Accept
FB 209 / 2007	Transverse Face Bend	19.0 x 3.0	No Discontinuity	Accept
RB 210 / 2007	Transverse Root Bend	19.0 x 3.0	No Discontinuity	Accept

[illegible]

Total :	4	Specimen(s)	☐ Attached Report =	Page(s)
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Certified by	  (Patoomporn Phongchotip / Project Engineer)
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Date : 05 MAR 2007